

## BASEL-BINNINGEN, Switzerland

WMO: 066010

Lat: 47.5411N

Lon: 7.5836E

Elev: 317

StdP: 97.57

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.3       | -5.3 | -11.6                         | 1.4 | -5.2 | -9.2 | 1.8 | -3.4 | 10.9                  | 11.2 | 9.3 | 8.9  | 2.2                   | 100  | 0.451 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 31.7            | 20.6 | 29.6 | 19.9 | 27.8 | 19.3 | 21.5                | 29.2 | 20.6 | 27.8 | 19.8 | 26.5 | 2.1                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.9                            | 14.3 | 24.3 | 18.1 | 13.6 | 23.2 | 17.5 | 13.0 | 22.6 | 63.8          | 29.5 | 60.7 | 27.9 | 58.0 | 26.7 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.2           | 6.4  | 5.3 | -10.2                      | 34.9 | 3.6                | 1.7 | -12.7                                              | 36.1 | -14.8      | 37.1 | -16.8      | 38.0 | -19.4      | 39.3 |    |    |
| (5)               |      |     | -10.7                      | 23.3 | 3.6                | 0.8 | -13.3                                              | 23.8 | -15.4      | 24.3 | -17.4      | 24.7 | -20.0      | 25.3 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.3        | 2.3   | 3.2   | 7.1   | 11.3  | 14.7  | 18.9  | 20.6  | 20.0  | 16.0  | 11.9  | 6.5   | 3.0   |  |  |
|                                                                        | DBStd       | 7.53        | 4.35  | 4.40  | 4.04  | 3.81  | 3.64  | 3.68  | 3.25  | 3.34  | 3.28  | 3.73  | 3.66  | 3.99  |  |  |
|                                                                        | HDD10.0     | 930         | 241   | 192   | 106   | 31    | 4     | 0     | 0     | 0     | 1     | 26    | 114   | 217   |  |  |
|                                                                        | HDD18.3     | 2787        | 497   | 424   | 348   | 212   | 122   | 37    | 14    | 20    | 85    | 201   | 354   | 474   |  |  |
|                                                                        | CDD10.0     | 1419        | 2     | 1     | 16    | 69    | 151   | 268   | 328   | 311   | 180   | 84    | 10    | 1     |  |  |
|                                                                        | CDD18.3     | 236         | 0     | 0     | 0     | 1     | 10    | 54    | 83    | 72    | 14    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 2018        | 0     | 0     | 0     | 20    | 110   | 472   | 738   | 579   | 94    | 5     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 642         | 0     | 0     | 0     | 1     | 19    | 149   | 258   | 203   | 13    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.1         | 2.6   | 2.6   | 2.5   | 2.1   | 2.0   | 1.9   | 1.9   | 1.7   | 1.6   | 1.8   | 2.2   | 2.6   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 775         | 49    | 51    | 50    | 60    | 79    | 89    | 74    | 103   | 54    | 47    | 71    | 46    |  |  |
|                                                                        | PrecMax     | 1069        | 95    | 153   | 102   | 117   | 130   | 184   | 135   | 181   | 152   | 106   | 178   | 84    |  |  |
|                                                                        | PrecMin     | 519         | 6     | 10    | 9     | 16    | 41    | 11    | 43    | 28    | 9     | 13    | 11    | 6     |  |  |
|                                                                        | PrecStd     | 132         | 24    | 32    | 22    | 28    | 25    | 39    | 24    | 43    | 34    | 26    | 40    | 21    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 14.4  | 15.4  | 20.7  | 25.8  | 29.3  | 33.2  | 34.4  | 34.6  | 28.4  | 23.6  | 17.5  | 14.1  |  |  |
|                                                                        |             | MCWB        | 10.4  | 8.9   | 11.9  | 14.9  | 18.8  | 21.3  | 21.3  | 20.8  | 18.6  | 16.2  | 12.4  | 10.0  |  |  |
|                                                                        | 2%          | DB          | 12.4  | 13.2  | 18.2  | 23.2  | 26.2  | 30.5  | 31.8  | 31.2  | 26.2  | 20.9  | 15.1  | 12.0  |  |  |
|                                                                        |             | MCWB        | 9.0   | 8.3   | 10.3  | 13.5  | 16.9  | 20.4  | 20.3  | 20.8  | 18.4  | 15.3  | 11.5  | 9.3   |  |  |
|                                                                        | 5%          | DB          | 10.3  | 11.4  | 15.9  | 20.8  | 24.1  | 28.3  | 29.5  | 28.8  | 23.9  | 18.9  | 13.2  | 10.2  |  |  |
|                                                                        |             | MCWB        | 7.6   | 7.5   | 9.8   | 12.9  | 15.8  | 19.7  | 19.8  | 19.9  | 17.5  | 14.6  | 10.4  | 7.8   |  |  |
|                                                                        | 10%         | DB          | 8.2   | 9.3   | 13.7  | 18.6  | 21.7  | 26.1  | 27.5  | 26.6  | 21.8  | 17.1  | 11.5  | 8.6   |  |  |
|                                                                        |             | MCWB        | 6.2   | 6.0   | 8.9   | 11.7  | 14.9  | 18.5  | 19.1  | 19.2  | 16.4  | 13.9  | 9.3   | 6.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.9  | 9.7   | 12.5  | 15.6  | 19.5  | 22.7  | 22.3  | 22.7  | 20.2  | 17.4  | 13.3  | 10.7  |  |  |
|                                                                        |             | MCDB        | 13.9  | 13.5  | 17.9  | 23.8  | 27.2  | 30.2  | 31.2  | 30.5  | 26.3  | 22.0  | 16.2  | 12.9  |  |  |
|                                                                        | 2%          | WB          | 9.4   | 8.7   | 11.2  | 14.3  | 18.1  | 21.2  | 21.2  | 21.3  | 19.1  | 16.2  | 11.9  | 9.4   |  |  |
|                                                                        |             | MCDB        | 12.0  | 12.7  | 16.1  | 21.3  | 23.7  | 28.5  | 29.0  | 28.8  | 24.5  | 19.4  | 14.4  | 11.7  |  |  |
|                                                                        | 5%          | WB          | 7.6   | 7.6   | 10.2  | 13.2  | 16.8  | 20.2  | 20.4  | 20.4  | 18.0  | 15.2  | 10.7  | 8.1   |  |  |
|                                                                        |             | MCDB        | 9.9   | 11.2  | 14.9  | 19.6  | 22.1  | 26.9  | 27.8  | 27.4  | 22.6  | 18.1  | 12.9  | 10.0  |  |  |
|                                                                        | 10%         | WB          | 6.3   | 6.3   | 9.2   | 12.1  | 15.6  | 19.2  | 19.6  | 19.6  | 17.0  | 14.1  | 9.6   | 6.8   |  |  |
|                                                                        |             | MCDB        | 8.3   | 9.0   | 13.4  | 17.6  | 20.4  | 24.9  | 26.3  | 25.6  | 20.8  | 16.8  | 11.3  | 8.5   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.1   | 6.6   | 8.9   | 10.1  | 9.7   | 10.0  | 10.2  | 9.8   | 9.2   | 7.6   | 5.6   | 5.0   |  |  |
|                                                                        |             | MCDBR       | 7.6   | 10.2  | 13.0  | 14.3  | 13.9  | 13.6  | 14.0  | 13.6  | 12.3  | 10.3  | 7.7   | 6.7   |  |  |
|                                                                        | 5% WB       | MCWBR       | 5.1   | 6.1   | 6.9   | 6.8   | 6.4   | 5.7   | 5.3   | 5.2   | 5.7   | 5.6   | 4.9   | 4.7   |  |  |
|                                                                        |             | MCWBR       | 7.1   | 9.5   | 11.2  | 12.8  | 11.9  | 12.1  | 12.2  | 12.3  | 10.7  | 9.2   | 7.1   | 6.2   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.308 | 0.334 | 0.368 | 0.405 | 0.415 | 0.419 | 0.418 | 0.411 | 0.388 | 0.365 | 0.330 | 0.305 |  |  |
|                                                                        |             | taud        | 2.446 | 2.369 | 2.311 | 2.229 | 2.248 | 2.278 | 2.274 | 2.307 | 2.352 | 2.401 | 2.449 | 2.449 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 775   | 824   | 846   | 846   | 850   | 847   | 841   | 831   | 816   | 776   | 738   | 731   |  |  |
|                                                                        |             | Edh at Noon | 66    | 88    | 109   | 130   | 132   | 129   | 128   | 119   | 104   | 85    | 66    | 59    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.07  | 1.88  | 3.12  | 4.34  | 5.05  | 5.94  | 5.67  | 4.93  | 3.73  | 2.33  | 1.23  | 0.86  |  |  |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.30  | 0.48  | 0.65  | 0.51  | 0.55  | 0.51  | 0.47  | 0.36  | 0.22  | 0.15  | 0.17  |  |  |

## Historical Trends

|      |                          | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|--------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                          |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only             | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (112 neighbors) | N/A   | N/A     | N/A    | +0.75   | +0.39 | +0.29 | N/A         | N/A     | +52     | +14     |

Nomenclature: See separate page